Revision | **R** Labs 2 & 4 | **File** 5 / 7

REV

Chapter 2 revision pack

حزمة المراجعة — القوانين، اختيار الرسم، الأخطاء، أوامر R، وخطة A+

1 · Formula and rule sheet

ITEM	RULE	REMEMBER
Class width طول الفئة	(H − L) / k → round UP	even 4.0 → 5; k is given
Next lower limit	$LCL_{i+1} = LCL_i + width$	LCL_1 = smallest value
Upper limit	$UCL_i = LCL_{i+1} - 1$	for whole-number data
Boundaries الحدود الفعلية	LCB = LCL − ½ unit UCB = UCL + ½ unit	±0.5, ±0.05, ±0.005 …
Width from a table	UCL − LCL + 1 = UCB − LCB = next LCL − LCL	not (UCB − LCB) + 1
Midpoint مركز الفئة	(LCL + UCL)/2 = (LCB + UCB)/2	x-axis of the polygon
Relative frequency	RF = f / n	ΣRF ≈ 1
Cumulative frequency	CF = running total of f	starts at 0, ends at n
Cumulative RF	CRF = CF / n	ends at 1
Percent	% = f / n × 100	categorical table, pie
Pie angle	degrees = f / n × 360°	angles sum to 360°

2 · Which graph?

DATA AND PURPOSE	GRAPH	X-AXIS
Quantitative, grouped: show the shape	histogram	class boundaries

DATA AND PURPOSE	GRAPH	X-AXIS
Quantitative, grouped: shape as a line; compare two distributions on one axis	frequency polygon	midpoints (+1 empty class each end)
Quantitative: how many values are below x	ogive	upper boundaries
Categories: compare frequencies	bar graph	categories (gaps)
Categories × groups	compound bar (grouped or stacked)	categories
Categories: parts of a whole	pie	—
Values over time: trend	time series	time
Small quantitative data set: shape and keeping every value	stem-and-leaf	stems

3 · Twelve traps

1. Rounding the width to the *nearest* whole number instead of **up**.
2. Leaving a whole-number width (4.0) unchanged.
3. Putting class **limits** under histogram bars instead of boundaries.
4. Leaving gaps between histogram bars, or no gaps in a bar graph.
5. Deleting a class with f = 0.
6. Plotting an ogive at midpoints or lower boundaries; it uses **upper** boundaries and starts at 0.
7. Forgetting the extra empty class at both ends of a frequency polygon.
8. Reading a stacked section as its top value instead of top − bottom.
9. Pie angles that don't total 360° (check!).
10. A stem-and-leaf plot with unordered leaves, a missing key, or a dropped empty stem.
11. Naming skew after the peak instead of the tail.
12. No title, unlabelled axes, or a missing zig-zag on an axis that doesn't start at 0.

4 · Removed this semester (recognise them only)

Pareto chart: a bar graph of a categorical variable with bars arranged from highest to lowest. **Dot plot**: each value plotted as a dot above a number line. **Back-to-back stem-and-leaf**: two data sets sharing one column of stems. **Misleading graphs**: a truncated y-axis or pictures scaled by area that exaggerate differences. These are marked "removed / not included" on the lecture slides and are absent from Dr. Amani's notes.

5 · R for Lab 2 (Data Basics) and Lab 4 (Tabulation, Graphs)

TASK	R
categorical frequency and percent	`t <- table(blood); t; prop.table(t)*100`

TASK	R
grouped frequency table with given classes	`table(cut(temp, breaks = seq(99.5, 134.5, by = 5)))`
cumulative frequency	`cumsum(table(cut(temp, breaks = seq(99.5, 134.5, 5))))`
histogram with the course boundaries	`hist(temp, breaks = seq(99.5, 134.5, 5), main = "Record highs", xlab = "°F")`
bar graph, horizontal bar graph	`barplot(table(drinks))` `barplot(table(drinks), horiz = TRUE)`
grouped and stacked bars from a matrix	`barplot(m, beside = TRUE, legend = TRUE)` `barplot(m)`
pie, time series, stem-and-leaf	`pie(table(x))` `plot(year, cost, type = "b")` `stem(x)`

R's `hist()` picks its own breaks unless you pass `breaks`, so always give the course boundaries. `stem()` may split stems automatically; use `stem(x, scale = 2)` to change that.

6 · A+ plan for Chapter 2

1. **Build the Example 2-2 table from the raw data**
   without notes: width, limits, boundaries, f, RF, CF, CRF. If yours matches, the procedure is learned.

2. **Draw the histogram, polygon and ogive**
   for the same table on graph paper, with title, axis labels and zig-zag. Memorise what goes on each x-axis.

3. **Read graphs back:**
   for each figure in the lessons, write one value you can read off it and one sentence of interpretation (CLO3).

4 **Textbook page 101, Q 5, 7, 19, 23**

(assigned in the outline), then the question bank, closed book.

خطة A+ بالعربي

(1) ابنِ جدول مثال 2-2 من البيانات الخام وحدك: الطول والحدود والحدود الفعلية والتكرارات النسبية والمتجمعة. (2) ارسم المدرج والمضلع والمنحنى المتجمع لنفس الجدول بالعناوين الكاملة، واحفظ ما يوضع على المحور الأفقي لكل رسم. (3) تدرّب على **القراءة**: من كل رسم استخرج قيمة واكتب جملة تفسير. (4) حل تمارين الكتاب صفحة 101 (5، 7، 19، 23) ثم بنك الأسئلة.

Chapter 2 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief · 01 · Lesson 2-1: Organizing Data · 02 · Lesson 2-2: Histograms, Frequency Polygons, Ogives · 03 · Lesson 2-3: Other Types of Graphs · **04 · Revision pack** · 05 · Exam-style questions · 06 · Full worked answers